

**MODEL PARAMETERS AND ITS SIZE FOR TRAINING AND MODEL
FILE**

For one IA Params:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
MATRIX	Key	4096	64	1	262144	1	MB
	Query	4096	64	1	262144	1	
	Horizontal	64	4096	1	262144	1	
	Vertical	64	4096	1	262144	1	
MLP	Hor	64	64	16	65536	0.25	MB
	Ver	64	64	16	65536	0.25	
Main Model Parameters and Size				Attention	1179648	4.5	GB
				MODEL:	2415919104	9	

Embedding	64
Layers (PAs)	8
Blocks	8
Attention Per layer (IAs)	32
Layers of MLP	16
Context	4096
Height	4096

1D
EV, EH, dv, dh, changeV, changeH, input, output expected

Incomplete Attention IA	Single Head
Partial Attention (Layer)	Consecutive IAs
Block or Complete Attention	Parallel PAs
Transformer	Multi or Single CA

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	KdotQ	4096	4096	1	16777216	64	MB
	Head	4096	4096	1	16777216	64	
MATRIX	Key	4096	64	1	262144	1	
	Query	4096	64	1	262144	1	
	Horizontal	64	4096	1	262144	1	
	Vertical	64	4096	1	262144	1	
MLP	Hor	64	64	16	65536	0.25	
	Ver	64	64	16	65536	0.25	
Continuous Calculation for every head	KEYS	4096	64	1	262144	1	
	QUERIES	4096	64	1	262144	1	
	Hlayers	64	16	2	2048	0.0078125	
	Activations	64	16	2	2048	0.0078125	
	gweights (hor)	64	64	16	65536	0.25	
	gweights (ver)	64	64	16	65536	0.25	
Complete Model Training Parameters and Size (one parallel is number of attention heads working in parallel)				Attention	35393536	135.015625	GB
				One Parallel:	283148288	1080.125	
				Model:	72485961728	270.03125	
Very Small	TOKENS	1	64	32768	2097152	8	MB
	1D	9	64	2048	1179648	4.5	MB
				72489238528	270.043457		GB

For Calculation:	NAME	DIMS			TOTAL VALUES	SIZE	UNIT
ATTENTION	KdotQ	4096	4096	1	16777216	64	MB
	Head	4096	4096	1	16777216	64	
MATRIX	Key	4096	64	1	262144	1	
	Query	4096	64	1	262144	1	
	Horizontal	64	4096	1	262144	1	
	Vertical	64	4096	1	262144	1	
MLP	Hor	64	64	16	65536	0.25	
	Ver	64	64	16	65536	0.25	
Continuous Calculation for every head	KEYS	4096	64	1	262144	1	
	QUERIES	4096	64	1	262144	1	
	Hlayers	64	16	2	2048	0.0078125	
	Activations	64	16	2	2048	0.0078125	
	gweights (hor)	64	64	16	65536	0.25	
	gweights (ver)	64	64	16	65536	0.25	
For Model Use Parameters and Size				Attention	35393536	135.015625	GB
				One Parallel:	283148288	1080.125	
				Model:	9060745216	33.75390625	
Very Small	TOKENS	1	64	32768	2097152	8	MB
	1D	9	64	2048	1179648	4.5	MB
				9064022016	33.76611328		GB